

From: [Thomasson, Alex](#)
To: [ostp-ai-rfi](#)
Subject: [External] AI Action Plan
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Dear folks,

Please accept the comments in the next few paragraphs in response to the Request for Information on the Development of an Artificial Intelligence (AI) Action Plan. These are recommendations focused on how AI can benefit agriculture in the U.S. They are well researched and come from a broad group of industry experts who worked together to write a larger issue paper on this subject, which will be published by CAST (Council for Agricultural Science and Technology) on March 31, 2025. Because I am submitting these recommendations prior to publication, please consider them to be attributable to me only — I don't want to have them attributed to the other experts or to CAST prior to publication, while the manuscript is still under review. I appreciate your consideration of them.

Best regards,

Alex Thomasson

J. Alex Thomasson, Ph.D., P.E.,
Professor, Department Head, and William B. and Sherry Berry Endowed Chair, Department of Agricultural and Biological Engineering
Founding Director, The Agricultural Autonomy Institute
Mississippi State University, P.O. Box 9632, Mississippi State, MS 39762

Comments on the Development of an AI Action Plan relative to Agriculture

AI has tremendous potential to benefit agriculture in terms of improved efficiency, precision, productivity, lifestyle improvement, etc. Efforts in and examples of generative AI as well as numerous other forms of AI for agriculture are proliferating. On the other hand, several barriers exist to realizing the full potential of AI in agriculture. These include the common incompatibility of agricultural data, the difficulty in making AI models widely applicable for agriculture, the lack of broadband connectivity in rural areas and remote farm fields in particular, concerns about lack of privacy in agricultural data, resistance to change among farmers and others in the agricultural supply chain, lack of a tech-savvy workforce, and high costs of adoption. While the development of AI technologies for agriculture grows, albeit slowed by the aforementioned barriers, concerns exist about societal implications of how it is used. These include the loss of agricultural knowledge by farmers and workers using AI, issues of cybersecurity, the needs of workers and consumers relative to AI in agriculture as well as other ethical issues, and finally the lack of adequate funding for research on AI in agriculture. On the other hand, the opportunities for the growth of AI in agriculture and its positive effects on the industry are immeasurable. To take advantage of these opportunities, it

is essential to promote innovation and competition in the industry, to advance American leadership in AI's development for agriculture, and to promote the development of standards that can support the growth of AI in agriculture while improving efficiency and reliability in safe and ethical ways.

Along these lines, we recommend that policy makers ponder the following initiatives, changes, and lines of thinking. It is important to contemplate regulations that would (a) create an enabling environment for AI in agriculture that aligns with agricultural practice and economic and environmental goals; (b) implement a self-regulatory approach among funding bodies, industry, academics, and farmers, similar to that of the 1975 Asilomar conference for biotechnology (Hurlbut, 2025); (c) enable farmers to easily engage with these systems and common data repositories; and (d) consider open-source tools as a means to increase competitiveness.

It is also important to ensure that adequate funding is available for (a) AI-in-agriculture research to catalyze discoveries and new applications and ensure U.S. competitiveness; (b) AI-in-agriculture research possibly in the form of additional AI institutes with a focus on agriculture, as envisioned by the National Artificial Intelligence Initiative (NAII); (c) broadband wireless communications infrastructure to enable farmers to participate in AI-driven innovations; and (d) public-private partnerships to create a viable financial ecosystem for innovation.

Furthermore, it is important to develop opportunities for education and training that (a) provide agricultural and technical, particularly AI-related, skills for the workforce of the future; (b) that equip extension agents, farmers, farm workers, and the agricultural workforce with the necessary skills to operate AI technologies; and (c) promote the decision-making benefits of precision agriculture and AI-driven technologies to farmers and agricultural supply chain members.

It is important to ensure that AI tools (a) are tailored to the specific needs of farmers to make the technology relatable and credible, and (b) engage workers in the design and implementation of the tools to ensure their perspectives are taken into account. Moreover, it is important to encourage transparency in how AI-driven decisions are made and recommendations are formulated – such as with TAI (trustworth AI) and XAI (explainable AI).

Additionally, it is important to consider a regulatory framework that promotes Ag Data Core Principles, which include (a) simple contracts between data providers and farmers, and (b) numerous protections for farmers.

It is also appropriate to consider regulations that ensure algorithms for AI in agriculture (a) are beneficial to large and small farms and those in high-income and low-income communities, (b) are developed with concepts like federated learning to decentralize data so as not to expose sensitive information, (c) are developed with diversified datasets so that the tools are

representative of a broad range of geographies, (d) are accessible and affordable, and (e) work with offline-compatible tools to help farmers in areas with limited internet connectivity.

Also, it is important to implement a comprehensive cybersecurity strategy involving academic, industry, government, and agricultural stakeholders to prevent unauthorized access, etc.

Finally, it is important to develop standards for AI technologies in agriculture to facilitate interoperability among brands of products as well as quality and safety assurance.

There is immense potential for AI to enable a next step change in agriculture. Expanding funding, training, etc. can enhance the adoption of AI in agriculture, supporting various sizes of farm operations through access to advanced AI tools and the ability to use them. Such an initiative would position the United States as a global leader in agricultural AI, driving economic growth, ensuring food security, and promoting sustainable practices.

Bibliography

Hurlbut, J. B. (2025). Taking responsibility: Asilomar and its legacy. *Science*, 387(6733), 468-472.

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